Design Brief – Review 2

0. Header

Project code	FA26SE260
Project name	CampusStay FPT – Dormitory Registration and Room-Scheduling Management System for FPT University's Tân Uyên Campus, Binh Duong
Team	CampusStay FPT — Lê Thành Minh Trung (Leader, SE183752), Nguyễn Xuân Lộc (SE183782), Trần Võ Quang Duy (SE182146), Võ Minh Luân (SE182794)
Supervisor	Thân Thị Ngọc Vân
Repository link (tag review-2)	https://github.com/tisou47/CampusStay-FPT-v0.1/tree/review-2
Version	1.1 (after Review 2 and the supervisor meeting of 05/11/2026; registration form 2.1)
Date	08/11/2026

1. Scope

1.1 Registration Changes

Section	Current	Proposed	Reason
Team, context, actors	No members; Bình Dương; Student, Staff, Admin	4 members; Tân Uyên ward, HCMC; basis: FPT dorms, VNU-HCM Zone B, Circular 27/2011; + Manager, Staff per Block; Family Visitor (external)	Review 1 and 2 feedback; 2025 merger
Room choice, contract, check-in/out	Select room/bed; digital contract; self check-in/out	Choose Room Type and Class, engine assigns; consent to rules; self check-in/out compared with a Staff baseline	Fairness; no e-signature; supervisor (05/11)
Payment, communication	VietQR payment; in-app, email, SMS	VietQR transfer + claim confirmed by a Manager; in-app, push, email	No bank API; SMS cost
Added scope	—	Term transition, registration-period requests, late check-in, Round 2, Utility and debt; requests in one module; Family Day; AI chatbot	Iteration 1; Review 2 and supervisor (05/11)
Integration, technology	SIS link; forecast; Flutter	Student list import + reconciliation; dashboards; Flutter app; local AI model	SIS not accessible; privacy

1.2 Out of Scope

Item	Reason
Room or Bed self-selection by students	The engine keeps gender and Block rules and fairness (D2)
Electronic contract signing	Needs a legal e-signature provider; consent to the house rules is recorded instead
SMS notifications	Cost; in-app, push and email cover the cases
Live SIS integration, bank API, forecasting, iOS app	No access or contract; list import, reconciliation, Manager confirmation, reports and the Flutter app (Android) instead
External AI APIs; ID numbers in visitor passes	Privacy (Law 91/2025/QH15): the chatbot runs on a local model; a pass holds a random token only
Visitors in rooms or overnight; parent accounts	Family Day covers common areas; Family Visitor has no account

2. System at a Glance

CampusStay FPT takes a student from sign-in to a Bed and through every Term of the stay: Applications by Room Type and Room Class, a fair batch assignment, VietQR payments confirmed by Managers, door codes, self-service check-in and check-out with a room-condition report checked against the Staff baseline, Term transitions, requests and maintenance in one module, monthly Utility billing, family events with

visitor check-in and check-out, and an AI chatbot for students and parents. Users: students (web and the Flutter mobile app), Staff of each Block, Managers and the Administrator. Core value: every Bed, payment and visit follows one set of configurable rules with an audit trail. Architecture: one Spring Boot API with PostgreSQL, a React web app, an Android app and a local AI model (Report 4 §1.1, Figure 1).

3. Key Design Decisions

ID	Decision	Options considered	Chosen option and rationale	Owner	Related FR / diagram
D1	System structure	Microservices; modular monolith	One Spring Boot API + one PostgreSQL, layered packages per feature: Allocation and payment in one ACID transaction; one API for three clients	Lê Thành Minh Trung	SDD §1.1–1.2
D2	How Beds are given	Pick a Bed first come; Staff by hand; batch engine	Students choose Room Type and Class; a deterministic engine places by priority and fit; a Manager approves per Block: fair, no race	Lê Thành Minh Trung	FR-ASG-01..03
D3	Renewal of residents	Re-apply each Term; auto-renew	Next-Term Allocation on the same Bed at M2; requests to M3, decisions by M4, fees M6, unpaid at M7 releases the Bed	Nguyễn Xuân Lộc	FR-ASG-07, 08; act-wf03
D4	Responsibility for furniture	Per Room only; per Bed only; shared items of the Room + personal items of each Bed	Items are Shared (Room) or Personal (one Bed); each student confirms own items at check-in; personal damage is charged to its student, shared damage as Staff decide	Trần Võ Quang Duy	F-16; FR-INV-03
D5	AI chatbot	Cloud LLM API; rule-based FAQ; local LLM with tools	A local model (Qwen via Ollama) calls read-only API functions limited to the asking user (public content only without sign-in): no data leaves the server, answers come from live data, not from the model's memory	Võ Minh Luân	F-21; architecture

4. Top Risks

ID	Risk	Impact	How we verify or mitigate	Owner
R1	Dormitory not operating yet: real operation may differ	Rework after opening	Requirements from FPT's running dorms, VNU-HCM Zone B process and comparison, Circular 27/2011; every price, window and rule configurable; weekly mentor review	Lê Thành Minh Trung
R2	Manual claim confirmation at the peak (≈ 3,000 Deposits in 48 h)	Students wait	Claim stops the clock; bulk confirm with count and total; hourly summary	Lê Thành Minh Trung
R3	One small VM (≈ 1 GB RAM): load peaks, and a local AI model needs several GB	Slow API or chatbot	NFR-01 tests; model on a separate host or small quantized and kept loaded; answers streamed; API unaffected if the chatbot is down	Võ Minh Luân

5. Known Gaps

ID	Gap	Location	Plan	Owner	Due
G1	Registration v1 items reworded (BM03)	Registration form	Form 2.1; self check-in/out kept (G8)	Lê Thành Minh Trung	08/11/2026
G2	Workflow tests TC-WF-01..06 are manual (SR-16)	Report 5 §4.2	Run on staging; automate in CI	Nguyễn Xuân Lộc	30/11/2026
G3	NFR-01, 04, 06 not measured yet; chatbot time and single check-in not specified (SR-07, SR-08)	Report 5 §4.4	Load script, chatbot timing, uptime check, run times	Nguyễn Xuân Lộc	15/11/2026
G4	No concurrency test for one Allocation per Bed per Term (SR-08)	TC-NFR-03	Parallel-commit test	Lê Thành Minh Trung	15/11/2026
G5	Mobile app has only a few automated tests and no manual test cases (BM02 1.0)	mobile/, Report 5	UI tests of key screens	Nguyễn Xuân Lộc	30/11/2026
G6	Statuses in the schema never stored (SR-11); Term never Closed (SR-12)	V4, V6, V7; SDD §3.1	Drop or use them; close Terms after the stay	Lê Thành Minh Trung	15/11/2026

ID	Gap	Location	Plan	Owner	Due
G7	Round 2 run by a Manager not built (BM02 1.0, High)	FR-APP-07	Closed 28/10/2026: Manager › Assignment › Round 2 (TC-APP-024)	Lê Thành Minh Trung	28/10/2026
G8	Self-service check-in/out: code has the Staff desk (SR-01, SR-15, High; SR-18)	F-08; ASG	Baseline, report, commitment, comparison	Lê Thành Minh Trung	30/11/2026
G9	Family Day (VIS) not built (SR-02, SR-15, High)	F-20; VIS	FR, UC, ERD, WF-06, API, screens, TC-FD	Võ Minh Luân	30/11/2026
G10	Chatbot: no FR, not in the repository or architecture (SR-03, High; SR-17)	F-21	Add the AI service, tools and tests; architecture without "planned"	Võ Minh Luân	30/11/2026
G11	Requests are separate features, not one module by type (SR-04)	F-19; REQ, MNT	Common request type and list; Report 3 and 4 updated	Nguyễn Xuân Lộc	30/11/2026
G12	Report 3 and 4 not yet updated to registration 2.1: UC, ERD, states, activity diagrams (SR-10, SR-14)	SRS, SDD	Redraw the activity diagrams with exceptions; RACI; Khu B comparison	Lê Thành Minh Trung	15/11/2026
G13	Shared and personal furniture not specified (SR-05)	F-16; INV	scope and bed_id; FR-INV-03	Trần Võ Quang Duy	30/11/2026

6. Proposed Mandatory Features

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-01	Sign-in (password, Google)	All	Valid credentials give a session; the 5th wrong password locks the account; a new @fpt.edu.vn Google account gets no password until it sets one; no token → 401, wrong role → 403	TC-F01-01..04	Done
F-02	Student accounts: verification and list import	Student, Manager	A pending account cannot apply; verify/reject notifies the student; an import previews each line and saves only valid lines	TC-F02-01..05	Done
F-03	Registration-period request and Term publication	Manager, Admin	A 2nd open request for a Term is refused; milestones out of order are named per field; publishing fixes the prices	TC-F03-01..05	Done
F-04	Apply for a Term	Student	Submit creates an Awaiting Deposit Application and a Deposit = Deposit % × Term price; a 2nd Application or stay length above the maximum is refused	TC-F04-01..05	Done
F-05	VietQR payment claim and confirmation	Student, Manager	A claim stops the deadline; a rejection needs a reason and returns the time left (≥ 12 h); a Paid or Void Invoice cannot be claimed	TC-F05-01..03	Done
F-06	Review and assignment	Manager	A rejection needs a note; the same input gives the same proposal; commit creates Allocations and Balances; Staff cannot run	TC-F06-01..05	Done
F-07	Balance, forfeit and door code release	Student, Staff, System	A Balance unpaid at its deadline is forfeited; the code is released only when paid, after check-in start and with a confirmed code	TC-F07-01..05	Done
F-08	Self-service check-in and check-out	Student, Staff, System	A check-in report is refused before the Balance is paid, without a Staff baseline or without the commitment; a report matching the baseline records the check-in at the submit time; a mismatch records nothing and opens a re-inspection; check-out deducts the Utility share and damage per item from the Utility Deposit	TC-F08-01..06	Planned
F-09	Late check-in	Student, Staff, Manager	A request after check-in end + max days is refused; approval moves the Balance deadline; rejection needs a reason	TC-F09-01..03	Done
F-10	Next-Term requests	Student, Staff	An approved change moves the next-Term Bed; a Room Class change needs all continuing roommates; extension without debt is automatic	TC-F10-01..06	Done
F-11	Requests window, reconciliation and next-Term fees	Manager, System	M2 creates next-Term Allocations once; M3 releases un-extended Beds; fees wait for decisions and reconciliation; M7 releases unpaid Beds	TC-F11-01..06	Done
F-12	Turnover and operational moves	Staff, Manager	A move reads both meters, charges nothing and asks a new code for the old Room; after fees Staff cannot move next-Term Beds	TC-F12-01..03	Done
F-13	Maintenance with routing, time limits and escalation	Student, Staff, Manager	A request gets the routed assignee and the Block deadline (urgent < normal); other Blocks' Staff do not see it; unassigned ones escalate	TC-F13-01..05	Done
F-14	Meter readings and Utility Invoices	Staff	A reading below the previous is refused; a CSV with a bad line saves nothing; closing issues Invoices split by days, rounded up	TC-F14-01..04	Done

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-15	Utility debt and termination	System, Manager	One overdue Utility Invoice gives warning 1 once; two overdue months after the final warning terminate the stay; past move-out it is Overdue	TC-F15-01..04	Done
F-16	Inventory, Room Class by Term, shared and personal furniture	Manager, Staff	A Room whose Beds are given keeps its class; only Managers set classes; each furniture item is Shared (Room) or Personal (Bed); a Personal item is confirmed by its own student at check-in	TC-F16-01..04	In progress
F-17	Notifications, announcements and push	All	Staff announce only to their Blocks; pinned announcements top the board; push after commit, dead tokens disabled	TC-F17-01..04	Done
F-18	Dashboards, reports, exports and search	Staff, Manager, Admin	Exports are CSV with the screen's columns and audited; search stays in the user's Blocks; Term dashboard shows milestones	TC-F18-01..04	Done
F-19	Requests in one module by type	Student, Staff, Manager	Every request type is created through one module with its type; the student sees all own requests in one list with one status set; a type not open now (e.g. swap outside M2–M3) is refused with the reason; Staff see only their Blocks' requests	TC-F19-01..03	Planned
F-20	Family Day management (VIS): events, registrations, visitor passes, visitor check-in/out	Manager, Student, Staff, Family Visitor (external), System	Publishing with invalid slots is refused; only residents register, in the period, within N visitors and slot capacity (also in parallel); one pass per approved visitor; a revoked, used or out-of-slot pass and a duplicate check-in are refused; ID checked before check-in; visitors not checked out are flagged	TC-F20-01..10	Planned
F-21	AI chatbot (local model, scoped data)	Student, Administrator; public mode without sign-in	Without signing in, the chatbot answers from public content only; a student asking for another student's Room or Invoice is refused; data in an answer comes only from read-only tools scoped to the session user; no call leaves the server to an external AI service	TC-F21-01..04	In progress

7. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method
NFR-01	Performance	p95 of 10 main read endpoints, 200 users, base data; chatbot first token, 5 parallel chats	≤ 1 s, errors < 1 %; chatbot ≤ 5 s	Load script on a server with production settings; timed chatbot script
NFR-02	Security	Permission and scope tests passing, incl. chatbot tool scope	100 %	Integration tests in CI
NFR-03	Integrity	Duplicate Allocations of a Bed in a Term; duplicate check-ins of an Allocation or of a visitor; Family Day slot over capacity	0	Unique indexes, row locks + concurrency tests
NFR-04	Availability	Health checks answering UP	≥ 99 % from 01/11/2026	Uptime monitor, every 5 min
NFR-05	Usability	Texts with a Vietnamese translation	100 %	MessagesTest + scan of web routes
NFR-06	Deployability	Push to healthy production; rollback	≤ 15 min; ≤ 5 min	Actions run times; timed rollback

8. Ownership Map

Member	Role	Owned parts (2–3)	Main workflow owned	Key decisions / PRs
Lê Thành Minh Trung (SE183752)	Leader, PM, BA, full-stack	Applications, payments (APP, BIL); assignment, residence, self-service check-in/out (ASG); architecture, DevOps	WF-01, WF-05	D1, D2; ADR-17, ADR-22
Nguyễn Xuân Lộc (SE183782)	Frontend developer, tester	Term transition (REQ next-Term, fees); requests module and maintenance (REQ, MNT); Flutter app and test documentation (Report 5)	WF-03, WF-04	D3; NFR-01
Trần Võ Quang Duy (SE182146)	Frontend developer	Term wizard (CFG + web); inventory, shared and personal furniture (INV); web UI, meters, i18n	WF-02	D4
Võ Minh Luân (SE182794)	Backend, AI developer	Family Day (VIS); AI chatbot (AI); sign-in, accounts and notifications backend (AUTH, NTF)	WF-06	D5

Appendix A – Defense 1.1 Targets

A1. Main Workflows

ID	Workflow	Actors	Trigger	Business outcome	Features	Entity states	Owner
WF-01	Registration to residence	Student, Manager, Staff, System	A student signs in during an open registration period	The student is a resident, checked in by the self-service report, with a door code	F-01, F-02, F-04, F-05, F-06, F-07, F-08, F-09, F-21	Application: Awaiting Deposit → Submitted → Approved → Allocated; Allocation: Assigned – unpaid → Paid → Resident; Check-in report: Submitted → Matched (or Mismatch → re-inspection)	Lê Thành Minh Trung
WF-02	Opening a registration period	Manager, Administrator, Student, System	A Manager requests a registration period	A published Term open to students	F-03, F-16, F-17	Registration-period request: Submitted → In progress → Done; Term: Draft → Published	Trần Võ Quang Duy
WF-03	Term transition of continuing residents	Student, Staff, Manager, System	M2 of the next Term	Continuing residents hold a paid Bed of the next Term; un-extended Beds are released	F-10, F-11, F-12	Next-Term Allocation: Continuing → Resident or Cancelled; Room-change / swap request: Submitted → Approved	Nguyễn Xuân Lộc
WF-04	Requests and maintenance with routing and time limits	Student, Staff, Manager, System	A resident submits a request (maintenance or another type)	The request is decided or the problem fixed and confirmed within the time limit, or escalated	F-19, F-13, F-17, F-16	Request: Submitted → Assigned → In Progress → Resolved → Closed (maintenance); Submitted → Approved / Rejected (other types)	Nguyễn Xuân Lộc
WF-05	Monthly Utility, debt and termination	Staff, Student, Manager, System	The month ends	Utility Invoices issued and collected; debtors warned or terminated	F-14, F-15, F-05, F-18	Invoice: Unpaid → Overdue → Paid; Termination: Notice sent → Checked out / Overdue	Lê Thành Minh Trung
WF-06	Family Day: visitor registration and check-in/out	Manager, Student, Staff, Family Visitor (external), System	A Manager creates and publishes a Family Day	Approved visitors enter and leave the common areas with their own passes; every entry and exit is logged; no-shows and visitors not checked out are resolved before the event is closed	F-20, F-17	Family Day: Draft → Published → Registration closed → In progress → Completed (or Cancelled); Registration: Pending → Approved / Rejected / Cancelled / Expired; Visit: Not arrived → Checked in → Checked out (or No-show)	Võ Minh Luân

A2. Demo Scenarios

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01	WF-01	Main	Stage A (registration open); se220001 eligible, no Application; Room baseline recorded by staffa01	1. se220001 applies (4-bed, Standard, stay length 2) and claims the Deposit transfer. 2. manager01 confirms the claim and approves the Application. 3. Stage B: se220004 (assigned, unpaid) claims the Balance; manager01 confirms. 4. staffa01 confirms the Room code; the release job runs. 5. se220004 fills the check-in report (personal and shared items, meter readings, photos), ticks the commitment and submits.	Application Submitted then Approved; se220004 is a resident (orange Bed) and sees the code; each step notified; the report matches the baseline and the check-in is recorded at the submit time	Lê Thành Minh Trung

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-02	WF-01	Exception	Stage A; se220003 has one rejected Deposit claim	1. manager01 rejects se220003's second claim with a reason.	The Deposit Invoice is Void and the Application Expired; the student is told the reason; the Bed count is unchanged	Lê Thành Minh Trung
DS-03	WF-02	Main	Stage 0; manager01, admin01	1. manager01 requests a period for the next Term and imports the excluded Rooms from the sample file. 2. admin01 builds the Term from the request, changes one price with a reason, completes the 7 steps and publishes.	Request Done and linked to the Term; manager01 sees the changed value; students see the registration card and the announcement	Trần Võ Quang Duy
DS-04	WF-02	Exception	Stage 0; a request for the same Term is open	1. manager02 files a second request for the same Term. 2. admin01 enters M3 before M2 in the Timeline step.	The second request is refused; the Timeline step names the wrong milestone; nothing is published	Trần Võ Quang Duy
DS-05	WF-03	Main	Stage A (requests window open); se210006 in A-505	1. se210006 asks for a room change. 2. staffa01 approves it with A-603 (stage B). 3. manager01 uploads the enrolled list and resolves groups 1–2. 4. manager01 issues the next-Term fees; se210001 pays and claims.	The next-Term Allocation moves to A-603; fees issued once with count and total; se210001 is Continuing and paid	Nguyễn Xuân Lộc
DS-06	WF-03	Exception	Stage C; se200005 did not pay the next-Term fee by M7	1. The M7 job runs. 2. staffa01 opens the turnover list.	The fee is Void, the next-Term Bed released, se200005 is on the leavers list with a check-out date	Nguyễn Xuân Lộc
DS-07	WF-04	Main	Stage A; se200009 in A-801, routing rules of Block A	1. se200009 reports an urgent air-conditioner fault with a photo and the item. 2. staffa02 (routed) starts and resolves it. 3. se200009 confirms.	Assigned at once with the urgent deadline; Closed; the item keeps a repair entry; notifications at each step	Nguyễn Xuân Lộc
DS-08	WF-04	Exception	Stage A; se200010 in A-802 has a request unassigned past its limit	1. The escalation job runs. 2. manager01 reassigns it.	One escalation in Overdue requests; after reassignment it closes once the request is back on track	Nguyễn Xuân Lộc
DS-09	WF-05	Main	Stage A; staffa01; meter sample file of Block A	1. staffa01 imports the readings and closes the month. 2. se200001 has an overdue Utility Invoice; the overdue job runs. 3. se200001 claims the transfer; manager01 confirms.	One Utility Invoice per resident split by days; warning 1 sent once; after payment the warning closes	Lê Thành Minh Trung
DS-10	WF-05	Exception	Stage A; se200004 terminated for Utility debt (A-703)	1. The move-out date passes without check-out; the job runs.	Termination Overdue; Managers and Block Staff told; A-703 needs a new door code	Lê Thành Minh Trung
DS-11	WF-06	Main	Stage A; Family Day on Sunday, slots 08:00–11:00 and 13:00–16:00, 100 visitors each, at most 3 per student; se200009 resident of A-801	1. manager01 publishes the Family Day. 2. se200009 registers father, mother and brother for slot 1; the registration is approved. 3. staffa01 checks the father's ID, checks him in and later out; the mother checks in only; the brother does not come. 4. manager01 has the open case resolved and closes the event.	3 separate passes; father Checked out, mother flagged at 11:00 as not checked out, brother No-show; the summary matches the visit log	Võ Minh Luân
DS-12	WF-06	Exception	Stage A; slot 1 is full; the father's pass has been used for check-in	1. se200010 registers a visitor for slot 1. 2. staffa01 scans the father's pass again.	The registration is refused (slot full, choose another slot); the second scan is refused and no duplicate check-in is recorded	Võ Minh Luân

A3. Functional Test Case Target

Source	Count	Minimum per item	Subtotal
Mandatory features	21	3	98
Main workflows (end-to-end)	6	1	6
State transitions of main entities	90	1	90
Agreed NFRs	6	1	6
Committed total			200

Automated in CI: 179 of 200 committed test cases (90 %); the rest are the end-to-end workflow cases, the manual state transitions and the measured NFRs (Report 5 §4). The chatbot is not a separate workflow: it changes no entity state and is covered by F-21 inside WF-01 (questions before and during registration). New cases (F-08, F-19, F-20, F-21, new transitions) are planned.